

Problem Definition

In the modern era, computer systems fail to detect and interpret human emotions in speech. Our group is looking to research how we can make a machine learning model that will allow accurate interpretation of human emotions in speech from a computer. Especially in the growing world of AI, this could have a profound impact on the development of new systems such as AI therapists where people can talk about their problems and won't receive non-human responses in return. The automatic identification of human emotions from speech signals will allow systems to make more empathetic responses and potentially be more effective in helping humans on an emotional level.

Data Collection

For our dataset, we will be using the CREMA-D audio dataset, which contains 7442 voice clips from 91 actors. We not only have access to a feature-extracted dataset containing each sample's gender, emotion, accent, etc, but we also have access to the raw audio files as well if we wish to go further. While researching, we did learn it is possible to read raw audio files in Python, so we might consider exploring this route.

Model Development

We are most likely going to need to extract a set of acoustic features to supplement the existing markup from the dataset. MFCCs seem to be the go to here, and we will feed that data into two complementary classifiers. A support vector machine which has good research behind it for this type of problem, and a Random Forest Ensemble which will provide feature importance scores, and help us understand which acoustic cues drive each emotion. Both models will undergo hyperparameter tuning via 5-fold cross validation to give us the most accurate and efficient configuration possible.

Evaluation and Validation:

In addition to standard evaluation metrics such as accuracy and precision, we will also track training time and prediction time for each model. These time-based metrics allow us to assess the computational efficiency of the models, which is essential for real-world deployment where both speed and resource use matter.

Specifically, we will measure:

- Training Time: The total time taken to train the model during each fold.
- Prediction Time: The average time taken to make predictions on the test data.
- Test Accuracy: The overall accuracy of the model on unseen data.

By considering both model performance and runtime efficiency, we aim to select a model that is not only accurate but also scalable and suitable for deployment in practical applications.

Each model will be trained and evaluated using k-fold cross-validation (with k=5 or 10) to ensure robust and generalized performance. This method helps mitigate overfitting by validating the model on multiple data splits, providing a more reliable estimate of its real-world effectiveness.